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Rydberg atoms

13.1 Introduction

Series of atomic spectral lines had been studied by emission from excited vapors even before Balmer described the regularity of the visible lines of hydrogen in 1884. However, direct excitation had been restricted to the few lowest-lying levels, particularly those accessible by electric dipole transitions from the ground state using classical light sources. The advent of tunable lasers in the 1970s changed such spectroscopic limitations enormously, allowing excitation to a very much larger number of states. Atoms in highly excited states, typically with principal quantum numbers $n \geq 10$, are called Rydberg atoms. They are oversized, fragile states of significant intrinsic interest. They are very rare in nature because of their size and fragility, but they are found in interstellar space (H I regions). One of the best sources for comprehensive descriptions of Rydberg atoms and references to the literature describing their exploration is Ref. [120]. It describes their characteristic properties and behavior both classically and quantum mechanically, and especially their strongly enhanced interactions with external fields.

The first thing to note is that Rydberg atoms are usually characterized by a single, highly excited electron while all the other electrons remain in their original states. Thus their properties are very nearly classical, and the atoms are much closer to the ideal hydrogen-like model of Chap. 7, along with appropriate quantum defects (see Chap. 10). The spectroscopy of such states is especially interesting because the active electron is much less perturbed by the other electrons in the atom than when it is in lower states. Moreover, there are some special circumstances in which the energy of a Rydberg state is degenerate with an atomic state having two electrons excited to lower levels, and such configurations yield further new information.

Second, Rydberg atoms do not usually decay back to the ground state by emission of characteristic radiation, so their study is mediated by other techniques that reveal new and interesting properties. Such properties arise because Rydberg atoms

have a huge dipole moment and very low binding energy resulting from the large average distance between the excited electron and the nucleus-plus-core of the rest of the atom with net charge e .

Third, transitions between Rydberg states are quite different from those connecting to the ground state for at least three reasons. One is that the energy differences are so small that they are in the radio or microwave frequency range. The resulting large wavelengths produce huge cross-sections (see Sec. 3.3.2 and other calculations of σ) that make Rydberg atoms exquisitely sensitive to very low levels of radiation. These can even include black body radiation at room temperature, the zero-point energy of the quantized field, and tiny resonant fields in cavity quantum electrodynamics experiments [121]. A second reason is that spontaneous emission is extraordinarily slow because of the small value of ω^3 in Eqs. (5.3) and (5.13) relative to that of optical transitions. This long lifetime against spontaneous decay, coupled with the huge absorption cross-section resulting from the large wavelength, makes Rydberg atoms into a wonderful laboratory for quantum information studies and quantum non-demolition demonstrations. Finally, the approximations such as those leading to Eqs. (3.4) and (13.2), and several others, have to be re-examined.

Fourth, gaseous samples of cooled and trapped atoms can be excited to Rydberg states quite efficiently, and their huge sizes enable a high collision probability. Since the binding energy is so low, a collision can ionize one of the partners while dropping the other one to a slightly lower Rydberg state. Thus interactions between atoms in a cold Rydberg gas can produce a cold neutral plasma whose parameters are unlike those of any plasmas previously studied. Moreover, the resulting slow electrons can readily recombine with the ions, and slow oscillations between the plasma state and the cold gas of neutral atoms state have been observed and studied. Thus Rydberg atoms provide access to plasma studies in new domains.

13.2 The Bohr model and quantum defects again

Many properties of Rydberg atoms can be understood correctly in terms of the relatively simple notions used to describe hydrogen atoms (see Chap. 7). For Rydberg atoms, Eq. (10.1) gives a very good approximation to the energy levels. The energies are negative and approach one another and $E_n \approx 0$ at large values of n . The energy separation between adjacent levels is $\Delta E_n = 2R_\infty/(n^*)^3$, the speed of the electron is $\alpha c/n^*$, and the radius of the orbit is $r_n = a_0(n^*)^2$. The picture that emerges of a Rydberg atom is one of an ion core and an isolated electron very far away, floating lazily around in a slow orbit, much like a distant planet of the solar system. For $n \sim 100$ the orbit is as large as a living cell and the electron speed is only 2×10^4 m/s.

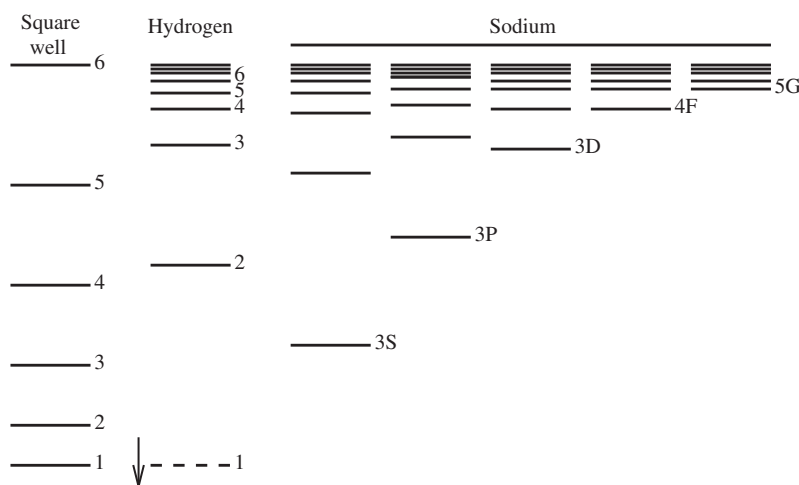


Figure 13.1 This shows the energy levels for various quantum mechanical systems. The left edge corresponds to a square well, beside it is the hydrogen atom, and the right part is the sodium atom. (Figure adapted from Ref. [122].)

Although the Bohr energy formula gives very good values for the hydrogen atom, it does not do as well as for other atoms because the valence electron's orbit may take it inside the closed-shell core where the potential descends very much more steeply than that of hydrogen (see Chap. 10). Such an electron therefore has much lower energy, as shown by the S and P levels in the central part of Fig. 13.1. The low angular momentum states (penetrating orbits) of alkali metals are therefore much more tightly bound than either the high angular momentum states (non-penetrating orbits) that see a nearly Coulombic potential or the corresponding n states of hydrogen. The differences between the high angular momentum states of alkalis and those of hydrogen may also arise because the ion core is composed of mobile electrons whose orbits can be perturbed by the valence electron (see Sec. 10.3).

These energy-level differences are characterized by the quantum defects as discussed and tabulated in Chap. 10. Equation (10.1) for the energies of the alkali energy levels agrees very well with spectroscopic data obtained from high-resolution studies of Rydberg atoms [123], and especially those with particularly simple cores such as helium and lithium [124]. These provide an excellent basis for comparison between atomic theory and experiment.

The electric field produced by the distant Rydberg electron at the ion core may not be uniform as is assumed in Sec. 10.3. Therefore the induced charge distribution need not correspond to a pure dipole moment, but may have quadrupole, octopole, and higher-order terms. For each of these there is an addition to the

Hamiltonian that can be treated similarly to the dipole term. It is a very curious property of the hydrogen radial wavefunctions that for all potentials $V(r)$ that vary as an inverse integer power of r of the form $1/r^k$ for $k > 1$, the leading term for the energy shift is proportional to $1/n^3$ [75, 125]. The result is that the quantum defect formula of Eq. (10.1) provides a very accurate prediction for the energy levels from interactions of different types (penetration as well as core polarization) because multipole terms are each proportional to some inverse power of r . Finally, neither the electron nor the core is at rest, resulting in various dynamic contributions to the polarizability (see discussion in Ref. [93]).

Equations (10.11) and (10.12) can be used to find approximate values for the quantum defects of non-penetrating states as given by Eq. (10.1), and the dependence is consistent with spectroscopic data as exhibited by the entries in Tab. 10.1 and Tab. 10.2. Therefore the tables of spectral data can be used to determine the dipole polarizabilities of the ion cores. The deviations from a perfect fit to equation Eq. (10.12) (or the unapproximated results of equation Eq. (10.11) can be used to determine the quadrupole polarizability or other higher-order effects (such as dynamic effects and retardation).

The discussion above has been restricted to Rydberg states of alkali-metal atoms. It is possible to study the Rydberg states of molecules or atoms with two or more valence electrons, such as alkaline earth atoms, and the added richness of the spectra provides new information. For example, after one electron is promoted to a Rydberg state, a second electron can be raised to one of the ion core's excited states. The atom now has too much internal energy to remain bound and autoionizes, emitting an "Auger electron" (see Sec. 14.5). This process may be enhanced by configuration interaction with another nearby state, and precise studies can be made. A theoretical structure called multichannel quantum defect theory (MQDT) has been developed to describe these processes [126] which can be measured carefully and cleanly by Rydberg spectroscopy.

13.3 Rydberg atoms in external fields

The description of Rydberg atoms in small external fields is very similar to that of ground-state atoms, but the field strengths that can be called small are very restricted. As highly excited electrons are weakly bound, moderate external fields can have large effects on them. For example, Sec. 13.3.1 below shows that the last term in Eq. (11.3) can usually be neglected, but also that it scales with n^4 . Thus modest fields can induce considerable diamagnetism in Rydberg atoms as a result of their large size.

The strength of the Coulomb field that binds the Rydberg electron to the parent ion scales as $r^{-2} \propto n^{-4}$, so for $n = 30$ it is only 6 kV/cm, a value easily achieved

Rydberg atom property	Magnetic field	Electric field
Linear term	Independent of n	$\propto n^2$
Quadratic term	$E^{(1)} \propto n^4$	$E^{(2)} \propto n^6$
Lin. term $\approx$ quad. term	$n = 18$ at $\mathcal{B}_{\text{lab}}$	$n = 60$ at $\mathcal{E}_{\text{lab}}$
Field ionization $\mathcal{E}_{\text{FI}}$	none	$\mathcal{E}_{\text{FI}} = \mathcal{E}_0/(2n)^4$
Ratio $E^{(2)}$ to ΔE_n , ΔE_{FS}	$\propto n^7$	$\propto n^9$
$E^{(2)} \sim \Delta E_n$	$n = 44$ at $\mathcal{B}_{\text{lab}}$	$n = 27$ at $\mathcal{E}_{\text{lab}}$
$E^{(2)} \sim \Delta E_{\text{FS}}$	$n = 9$ at $\mathcal{B}_{\text{lab}}$	$n = 6$ at $\mathcal{E}_{\text{lab}}$
Selection rules for diagonal matrix elements	$\begin{cases} \Delta \ell = \pm 2, 0 \\ \Delta m = 0 \\ \Delta n \text{ none} \end{cases}$	$\begin{cases} \Delta \ell = \pm 1 \\ \Delta m = 0 \\ \Delta n \text{ none} \end{cases}$

Table 13.1 A summary of a few of the field-dependent properties of Rydberg atoms that are discussed in the text. The quantities used for numerical examples are $\mathcal{E}_{\text{lab}} = 1 \text{ kV/cm} \approx 2 \times 10^{-7} \mathcal{E}_{\text{au}}$ and $\mathcal{B}_{\text{lab}} = 1 \text{ T} \approx 4 \times 10^{-6} \mathcal{B}_{\text{au}}$. Lin., linear; quad., quadratic.

in the laboratory. Under such conditions, any approximation of the effects of an applied field as being small compared with the basic atomic structure is destined for failure. The energy-level scheme of Rydberg atoms in electric fields having even a few percent of this size looks very different from the familiar zero-field scheme, and in fact, the Stark shifts can easily exceed the separation between levels in adjacent n -manifolds. The discussion below describes some of these surprising features.

The scaling of various energies of Rydberg atoms with n determines many of their characteristics. However, it is important to note that, even though $1/n^3$ scaling applies to both the interval between n -manifolds of hydrogen given by $\Delta E_n = 2R_\infty/n^3$, and the fine-structure interval $\Delta E_{\text{SO}} = \alpha^2 R_\infty/(2n^3)$ as given in Eq. (8.23) for hydrogen, their magnitudes differ by a factor of α^2 . Nevertheless, this strong n -dependence has important consequences for Rydberg atoms as summarized in Tab. 13.1.

13.3.1 Rydberg atoms in magnetic fields

The description of the Zeeman effect for Rydberg atoms begins with the second term of Eq. (11.3). It uses eigenfunctions that are the same as those of the field-free case with Hamiltonian $\mathcal{H}_0$ because the interaction term there becomes $g\mu_B \mathcal{B}L_z$. Since this commutes with both the ∇^2 term and $V = V(r)$ terms of the Hamiltonian $\mathcal{H}_0$, there are indeed simultaneous eigenfunctions of L_z and $\mathcal{H}_0$ and thus the Schrödinger equation can still be separated in spherical coordinates into solvable radial and angular parts.

The quadratic term of Eq. (11.3) that does not commute with $\mathcal{H}_0$ is given by

$$\mathcal{H}' = \frac{e^2}{2m} |\vec{\mathcal{A}}|^2 = \frac{e^2 \mathcal{B}^2 r^2 \sin^2 \theta}{8m}, \quad (13.1)$$

where in the last step it is assumed that the magnetic field is in the z -direction, as in Chap. 11. This term is generally much smaller than the linear term, as can be seen by estimating their ratio. The linear term is of the order $\mu_B \mathcal{B}$, whereas for the quadratic term the expectation value of the r^2 factor can be found from Tab. 7.6 using $\ell \ll n$. Assuming the $\sin^2$ term is of order unity, the ratio then becomes

$$\frac{(e^2/2m) |\vec{\mathcal{A}}|^2}{(e/m) \vec{\mathcal{A}} \cdot \vec{p}} \sim \frac{e^2 \mathcal{B}^2 \langle r^2 \rangle / m}{\mu_B \mathcal{B}} = \frac{2n^4}{Z^2} \frac{\mathcal{B}}{\mathcal{B}_{\text{au}}}, \quad (13.2)$$

where $\mathcal{B}_{\text{au}} \equiv R_\infty/\mu_B \approx 2.35 \times 10^5$ T is the atomic unit of magnetic field. Since typical laboratory magnetic fields do not exceed a few T, in general this ratio is $\ll 1$. However, for atoms in highly excited states with large n or large fields, the second-order shift becomes comparable to or larger than the first-order shift and thus it is evaluated below (see Ref. [127]).

Since the quadratic Zeeman effect term of Eq. (13.1) is generally small compared with the other terms of the Hamiltonian of Chap. 11, its energy shift can be calculated by treating it as a perturbation. Using $\mathcal{H}'$ of Eq. (13.1) with the hydrogenic wavefunctions, the first-order energy shift $\Delta E^{(1)}$ is

$$\begin{aligned} E^{(1)} &= \langle \psi_{n\ell m} | \mathcal{H}' | \psi_{n\ell m} \rangle \\ &= \frac{e^2 \mathcal{B}^2}{8m} \langle R(r) | r^2 | R(r) \rangle \langle Y_{\ell,m}(\theta, \phi) | \sin^2 \theta | Y_{\ell,m}(\theta, \phi) \rangle \end{aligned} \quad (13.3)$$

Even though this term is proportional to $\mathcal{B}^2$, it is the result of a first-order perturbation calculation as indicated by the label $E^{(1)}$.

The radial integral above is found from Tab. 7.6. For the angular part, one can use the relation $\sin^2 \theta \propto (Y_{2,0} + bY_{0,0})$, where b is a constant, so the angular integral becomes $\langle Y_{\ell m} | (Y_{2,0} + bY_{0,0}) | Y_{\ell m} \rangle$. Since a product of two spherical harmonics can be re-written as the sum of two spherical harmonics as given in App. C.5, the angular part is also readily evaluated. The final result is

$$\begin{aligned} E^{(1)} &= \frac{n^2 a_0^2 e^2 \mathcal{B}^2}{16mZ^2} (5n^2 - 3\ell(\ell+1) + 1) \left(\frac{\ell^2 + \ell - 1 + m^2}{(2\ell-1)(2\ell+3)} \right) \\ &\approx \frac{n^4 R_\infty}{3Z^2} \left(\frac{\mathcal{B}}{\mathcal{B}_{\text{au}}} \right)^2 \end{aligned} \quad (13.4)$$

for $n \gg \ell$.

Atomic diamagnetism scales with n^4 as in Eq. (13.4) for the case of the large n considered here. The ratios of $E^{(1)}$ to ΔE_n and to ΔE_{FS} both scale as n^7 , so that $E^{(1)} \sim \Delta E_n$ for $n \sim 44$ and $E^{(1)} \sim \Delta E_{\text{FS}}$ for $n \sim 9$, both for $\mathcal{B} \sim 1$ T (see Tab. 13.1). Moreover, for $n \sim 18$ the diamagnetic energy is comparable to the Zeeman energy at the same 1 T field. Some spectroscopic measurements were reported in Ref. [128].

More accurate calculations of diamagnetic energy shifts require the off-diagonal matrix elements of the $\mathcal{B}^2$ part of the Hamiltonian. It is clear from the $\sin^2 \theta$ dependence that the only non-vanishing angular integrals are those for which $\Delta \ell = \pm 2, 0$: $\mathcal{B}^2$ can only connect states of the same parity and must satisfy a “triangle condition” on the ℓ values. It is also clear that $\Delta m = 0$ and that there is no selection rule on n . For strong fields and/or large values of n , perturbative calculations are not satisfactory and other methods of calculations are needed.

Some very interesting results were reported on the diamagnetic spectra of barium and lithium. A laser pulse tuned to excite Ba atoms to a low-lying excited state was followed by a broad-spectrum light pulse whose absorption spectrum was measured in a field of $\mathcal{B} \approx 2$ T [129]. It revealed a series of transitions whose frequencies corresponded well with the expected results. A similar experiment with Li presented confusing results [130] that were later interpreted as the effect of the electric field seen by the atoms as a result of their motion in the magnetic field. The effect was much more pronounced in Li because its smaller mass resulted in a higher thermal speed.

13.3.2 Rydberg atoms in electric fields

A quantum mechanical description of Rydberg atoms in electric fields begins by separating the Schrödinger equation in parabolic coordinates as done in Sec. 12.3. In contrast to the weak magnetic field case where the zero field eigenfunctions are also eigenfunctions of $\vec{p} \cdot \vec{\mathcal{A}}$, the resulting radial differential equation for an electric field is not exactly solvable. Nevertheless, having the angular part separated makes the energy and eigenfunction calculations much easier. The lowest-order energy shifts are given in Eqs. (12.17) and (12.19), which scale as n^2 for the first order and n^6 for the second order [75]. With such strong dependence on n it is easy to see why Rydberg atoms are so very sensitive to electric fields. One can do higher-order perturbation theory [119], diagonalization with a truncated basis set [115], or other approximate calculations [75]. The definitive solution for many cases of practical importance was given in Ref. [118].

13.3.3 Energy estimates and quantum defects

In the case of applied magnetic fields, the relevant term in the Hamiltonian derived from the canonical momentum of the (atom plus field) system by replacing the

vector potential $\vec{\mathcal{A}}$ by $\vec{p} + e\vec{\mathcal{A}}$ in Eq. (11.1). By contrast, for electric fields the Hamiltonian term comes from the scalar potential term Φ , and is simply $e\mathcal{E}z$ for $\vec{\mathcal{E}}$ in the z -direction as in Eq. (12.1). This term connects states having ℓ -values differing by ± 1 , and for a particular n -manifold their zero-field energies are given in terms of a quantum defect δ_ℓ (see Chap. 10). The energy shift caused by the electric field can be found by evaluating $\mathcal{H}' = e\mathcal{E}z$ in this basis using the non-degenerate energies $E_{n,\ell}$. Consider the case where two of these energies $E_{n,\ell}$ and $E_{n,\ell'}$ are much closer together than all other energies. Since these are states of definite parity (eigenfunctions of $\mathcal{P}$), the diagonal elements of $\mathcal{H}'$ are zero. Denoting the off-diagonal element between these two nearest states as $C = \langle \psi_{n\ell m} | \mathcal{H}' | \psi_{n\ell' m} \rangle$, the Hamiltonian matrix becomes

$$\mathcal{H} = \begin{pmatrix} E_{n,\ell} & C \\ C & E_{n,\ell'} \end{pmatrix}. \quad (13.5)$$

Diagonalizing this Hamiltonian matrix yields eigenvalues $E'_{n,\ell}$ and $E'_{n,\ell'}$

$$E'_{n,\ell;n,\ell'} = 1/2(E_{n,\ell} + E_{n,\ell'}) \pm \sqrt{C^2 + 1/4(E_{n,\ell} - E_{n,\ell'})^2} \quad (13.6)$$

where the $+$ ($-$) sign is for the state n, ℓ (n, ℓ'). For the two states considered these energies are exact, but in general other nearby states that are not taken into account make an additional contribution to the energies.

One can take an estimated value of $C \sim e\mathcal{E}a_0n^2$ that is consistent with a classical result because a_0n^2 is simply the average separation between the electron and the core. In the limit of small $\mathcal{E}$, Eq. (13.6) can be expanded to find the shift (again $+$ ($-$) sign is for the state n, ℓ (n, ℓ'))

$$E_{n\ell;n,\ell'}^{(1)} \approx \pm \frac{e^2\mathcal{E}^2a_0^2n^7}{(\delta_{\ell'} - \delta_\ell)R_\infty}. \quad (13.7)$$

This is quite different from the second-order perturbation calculation of Eq. (12.19) because this is a first-order calculation on a basis of only two states, and the $\mathcal{E}^2$ dependence arises from expanding the radical of Eq. (13.6). It is to be emphasized that the shift $E^{(1)}$ scales as n^7 . Note that if $E_{n,\ell} < E_{n,\ell'}$ the state ℓ shifts downwards and the state ℓ' shifts upwards, since $\delta_\ell > \delta_{\ell'}$. The states seem to repel each other, which is generally true in quantum mechanics, and this leads to the non-crossing of the states (see App. 11.C).

The Zeeman effect has no such spectacular enhancement. The magnetic shift of atomic energy levels is proportional to the Landé g -factor which depends only on the spin and the angular part of the wavefunction $Y_{\ell,m}(\theta, \phi)$, and varies by no more than a factor of three over almost all atomic states.

13.3.4 Numerical calculations

An extension of this kind of approximation is to calculate a larger submatrix of the infinite Hamiltonian and diagonalize it. This was done very successfully in the early work on the Stark effect in Rydberg states (see Ref. [131]). Because of the selection rules as discussed in the beginning of Sec. 12.2 one has to take into account only the coupling between states where $\Delta\ell = \pm 1$ and $\Delta m = 0$. Since the energy shifts are very large and scale with a high power of n (*i.e.*, see Eq. (13.7)), and the separation between the principal manifolds that scales with $1/n^3$ becomes small at high n , modest electric fields can strongly split these levels, can thoroughly mix the ℓ states so that ℓ is no longer a good quantum number, and can bring together states of different n . Of course, these states cannot cross, according to App. 11.C, and so the multitudinous number of sublevels quickly develop into a tangled set of loci. The principle is illustrated in Fig. 13.2. This energy-field region is called the “spaghetti region”.

In some cases the quantum defects may even place states of higher n with small ℓ between manifolds of $n - 1$ and $n - 2$ so there is quite a complicated level structure. Adding this feature to the already strongly mixed ℓ states makes a truly complicated energy-level diagram worthy of the name “spaghetti region”. There are many such diagrams in the literature of Rydberg spectroscopy (see Ref. [115]). There is no such enhancement of the Zeeman splitting at high n because the magnetic moment

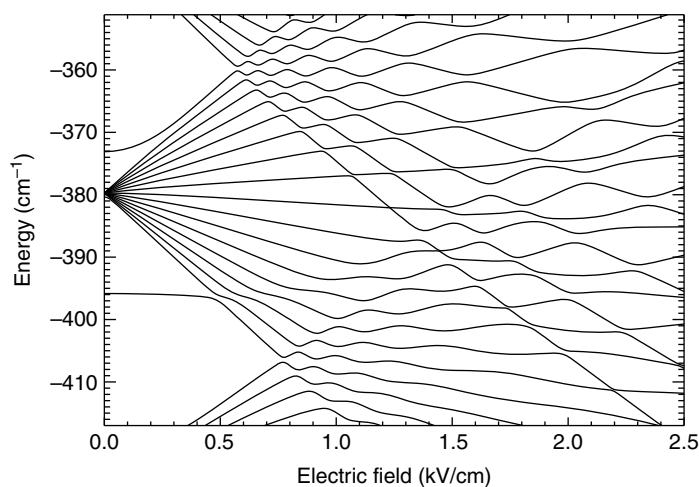


Figure 13.2 The spaghetti region begins at quite low fields when states of $\ell \sim n$ begin to split strongly to the point where states of different ℓ are brought together and strongly mixed. These do not cross, and the large multitude of states makes several sequential anti-crossings, resulting in a complicated diagram. Here the calculation for the $n = 16$ state of Na is shown for $m = 0$.

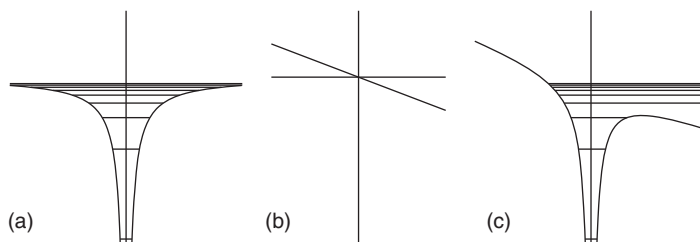


Figure 13.3 Part (a) shows the Bohr levels in a Coulomb potential and part (b) shows the potential of a uniform electric field. Part (c) shows that the sum of these has unbound states degenerate with each bound state. The peak on the right side is actually a saddle-point in three dimensions.

does not depend on the distance between electron and nucleus as does the electric dipole moment.

13.3.5 Field ionization

A simple, semi-classical description of the effects of a dc electric field $\mathcal{E}$ on Rydberg atoms shows that there are no perfectly bound states even for very weak fields. For a field in the z -direction there is always a finite distance at large enough positive z from the atom where the potential energy $e\mathcal{E}z$ is lower than the electron's binding energy (see Fig. 13.3). Moreover, the result of this large electron–nucleus separation of the Rydberg states makes them extremely sensitive to modest electric fields because of their high polarizability. Some of these large responses to electric fields are also summarized in Tab. 13.1.

Because of their relatively small binding energy, Rydberg atoms can be readily ionized by electric fields. The potential for a Rydberg electron in the field $\mathcal{E}$ can be written as $V = -e/4\pi\epsilon_0 r - \mathcal{E}z$, and it has a local maximum value $V_{\max} = -2\sqrt{e\mathcal{E}/4\pi\epsilon_0}$ at $z = \sqrt{e/4\pi\epsilon_0\mathcal{E}}$ (see Fig. 13.3). Clearly an electron with energy $E_n > eV_{\max}$ can escape. The condition for field ionization becomes

$$\mathcal{E}_{FI} > \frac{e}{4\pi\epsilon_0 a_0^2} \frac{1}{16n^4} \equiv \frac{\mathcal{E}_{\text{au}}}{16n^4} \quad (13.8)$$

where $\mathcal{E}_{\text{au}} \equiv e/(4\pi\epsilon_0 a_0^2) = 2R_{\infty}/ea_0 \approx 5.1 \times 10^{11}$ V/m is the atomic unit of electric field. For $n = 15$, field ionization occurs at $\mathcal{E} \approx 6$ kV/cm, a readily attainable laboratory field.

The model used to describe the field ionization above is rather simplified. One reason is that some electrons occupy states with wavefunctions concentrated away from the saddle-point produced in the three-dimensional potential by the applied electric field. They may have enough energy to field ionize, but do not sample

the potential at the escape route. Another arises because electrons with insufficient energy to ionize classically may escape by tunneling under the barrier. Further complications arise from the mixing of the wavefunctions by the field, resulting in excited states that are superpositions of zero-field eigenfunctions. Field ionization is a separate topic of serious investigation [132].

13.4 Experimental description

The apparatus used for Rydberg experiments can be simple for many atoms, especially the alkalis. A small stainless steel oven is loaded with a few grams of the metal and heated to a few hundred degrees. The oven is mounted in a modest vacuum system ($P \sim 10^{-6}$ torr) of characteristic linear dimension of 0.5 m resulting in a thermal atomic beam. Although some metals require a heat pipe oven, and some Rydberg spectroscopy has been done on rare gases that require a discharge for excitation but no oven, the system described above is the most common. The atomic beam is collimated by a series of apertures or by transverse laser cooling (see Chap. 19) and crossed by one or more carefully tuned laser beams that enter the vacuum system through appropriately placed windows (see Fig. 13.4a).

Population of the Rydberg states is generally done by laser excitation in two steps. Figure 15.2 shows that the ionization potential of most atoms is several eV, and is approximately 5 eV for all the alkalis. Since this corresponds to UV light, it is easier to populate the Rydberg states that are near the ionization limit with two steps as shown in Fig. 13.4b, using visible light from readily available tunable lasers. Selection rules then require $\Delta\ell = 0$ or 2.

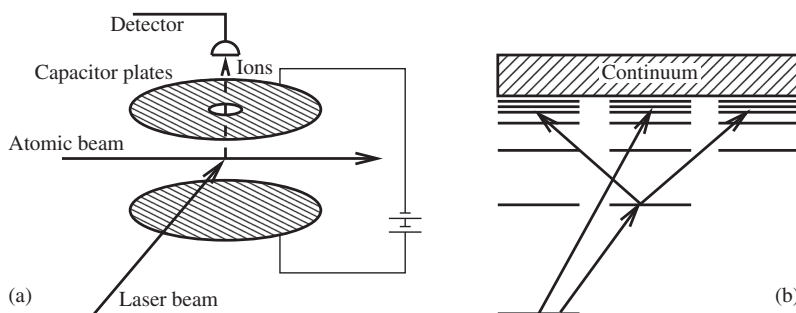


Figure 13.4 Part (a) is a schematic diagram of a typical setup for producing a beam of Rydberg atoms. The atomic beam is crossed by one or more laser beams between two metal plates that apply the electric field. One of them has a hole through which ions produced by field ionization can pass to the detector. Part (b) shows a typical two-step excitation scheme to move atoms to energies ~ 5 eV above their ground state, using visible light that excites the appropriate transitions.

At first it might seem that detection of Rydberg states could be very much the same as detection of other excited atomic states: one simply looks at the radiation emitted when the atom decays to its ground state. More careful consideration reveals that Rydberg atoms do not decay very well by radiation, and in fact may have lifetimes long enough for the atoms to hit the wall at the far end of the apparatus without ever emitting light.

Transitions down to adjacent Rydberg levels Δn away are very weak and slow because their frequencies are very low ($\Delta E_n = 2R_\infty \Delta n/n^3$) and spontaneous transition probabilities are proportional to the cube of the frequency ($1/n^9$). Transitions down to low-lying or ground states that have much higher frequencies are very weak because the radial wavefunctions of Rydberg levels have very many oscillations and the integral of such oscillatory functions multiplied by one with only very few nodes is a small fraction of a_0 . The result is that detection of Rydberg atoms by spontaneously emitted radiation is ineffective.

The usual procedure is to mount flat plate electrodes on either side of the atomic beam and apply a sufficiently strong dc electric field to ionize the atoms (see Fig. 13.4a). Because the Rydberg electron is far away from the nucleus and core, it can easily be pulled away by the applied field and detected. The laser beams are tuned to the transitions of interest and directed into the apparatus where they cross the atomic beam between the field plates. The optical excitation is performed with the desired voltage on the field plates, and shortly afterward the voltage is stepped up sufficiently to ionize a selected n -state ($\mathcal{E}_{\text{FI}} > \mathcal{E}_0/16n^4$). The negatively charged plate has an array of small holes, and many of the ions accelerated toward it pass through the holes to a detector mounted on the opposite side. Such field ionization, as described earlier, is very efficient for Rydberg atom detection.

13.5 Some results of Rydberg spectroscopy

When atoms are subjected to extremely strong electric or magnetic fields, that is, those fields that produce forces comparable to the Coulomb binding force, their structure is markedly changed. When the external field is so large that it can no longer be considered a perturbation, the ordinary solutions to the Schrödinger equation are not even approximately right. Theoretical studies of atoms in such conditions have been stimulated by interest in plasma and astrophysical problems, but experimental measurements have not been done because laboratory fields cannot be made large enough [133].

Rydberg atoms have provided a testing ground for the theories of atoms in strong fields because laboratory fields can easily be made to satisfy the strong field condition. The signals are typically studied as a function of laser frequency, applied dc electric [115] or magnetic fields [123, 134, 135], applied rf or microwave fields

that induce transitions between fine-structure levels or to other Rydberg levels [128, 130], background gas pressure, or combinations of these and other influences.

13.5.1 Stark spectroscopy

Soon after the advent of tunable lasers the spectroscopy of Rydberg atoms in weak and strong fields became a subject of great interest. Figure 13.5 shows signals resulting from step-wise excitation of Na as in Fig. 13.4b. Each vertical trace represents a laser frequency scan at a given electric field much smaller than the ionizing field [131]. A pulsed yellow laser excited atoms into the 3P level between the electric field plates and a pulsed violet laser subsequently populated the Rydberg state when its frequency was resonant with the transition. A few μs later the applied electric field was pulsed to a high enough value to ionize the Rydberg states, producing the signal. The measured Stark spectrum of the sodium atom is strikingly laid out in Fig. 13.5, and agrees well with the calculations, as shown in Fig. 13.2.

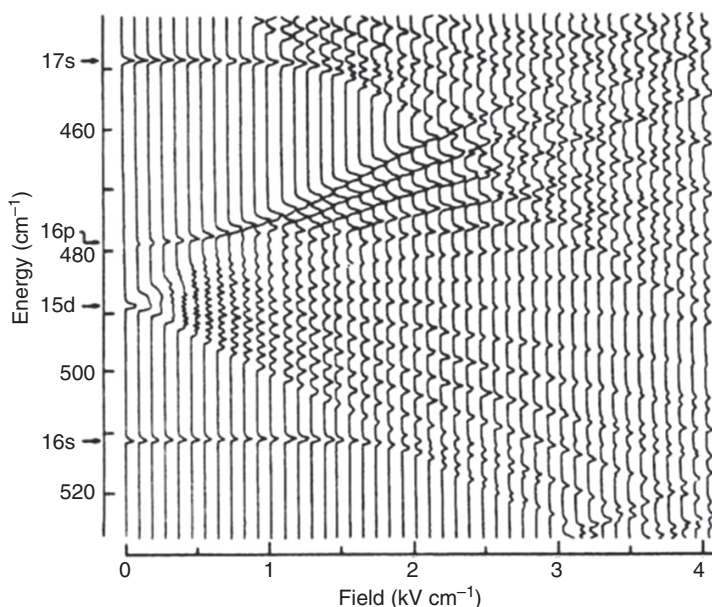


Figure 13.5 Each vertical trace in this figure is a plot of the ion current from a laser frequency scan at a given electric field much smaller than the ionizing field (horizontal axis). A few μs after the laser pulses that excited the atom through an intermediate state, the applied electric field was pulsed to a high enough value to ionize the Rydberg states, producing the signal. In this case the first laser pulse (yellow) excited Na from its ground 3S state to its first excited 3P state, and a second (blue) laser pulse, whose frequency was scanned, populated the Rydberg states. (Figure from Ref. [131])

One of the interesting features of Fig. 13.5 can be seen at low fields just above $E = -480 \text{ cm}^{-1}$. Selection rules forbid excitation of the P states from the 3P state as shown on the trace for zero field, but even the next trace shows signs of a peak at $\sim 100 \text{ V/cm}$, and there is a clear, strong peak at a few hundred V/cm. The transition is allowed because the electric field mixes the $n = 15$ states with higher angular momentum into the 16P state, and the resulting D-state component is readily excited from the 3P state. The curved locus of the eigenenergy plot soon straightens out as a result of the strong mixing that causes a linear Stark shift, as shown in Fig. 13.2.

Spectra similar to that of Fig. 13.5 can also be obtained at field values higher than that of Eq. (13.8) [132]. At first this may seem surprising because the ionization rate is so high that the states should be broadened into the photoionization continuum. However, it must be remembered that the total orbital angular momentum ℓ is no longer conserved in such a strong field so that every n state contains a mixture of higher and lower ℓ -values up to $\ell = n - 1$. Some of these have eccentric orbits that stay very close to the nucleus on the downfield side so that they do not sample the saddle-point area that would permit their escape. Thus their energy is above the limit corresponding to Eq. (13.8) but they are still quite well bound, so they result in narrow spectral features. By contrast, the rapidly ionizing orbital states are so broadened that the excitation to them is not noticeable. Observation of these resonant excitations does not require a delayed ionizing pulse, but instead the detector is gated to record only events that happen within a few μs after the laser pulse. This scheme is complementary to the data acquisition method of Fig. 13.5.

13.5.2 Precision measurements on high- ℓ states

Rydberg atoms are exquisitely sensitive to small effects, and for that reason are ideal for exploring them. One example is the use of precision microwave spectroscopy between neighboring Rydberg sublevels to measure retardation in the interaction between core electrons and the excited one [136]. Helium is chosen for these high-precision measurements because the wavefunction of the one-electron core can be known to extraordinary accuracy.

The excitation scheme is quite different. Ions from a discharge are accelerated through $\sim 10 \text{ kV}$ and then impinge on a thin foil where the violent interactions excite very many different states. Among these are those in the $n = 10$ manifold, and these are readily excited to higher n -values by a CO_2 laser at $\lambda \approx 10 \mu\text{m}$ for easy detection. The resolution of the process easily distinguishes among the $n = 10$ sublevels so that microwave transitions among them are readily detected. For states with $\ell > 5$, the electron wavefunction is so well separated from the core by the centrifugal potential of Eq. (7.9b) that spectroscopy is sensitive to effects

normally neglected, such as retardation. In Ref. [136] the reported precision was better than 2 kHz out of 500 MHz. Systematic effects such as Stark shifts from random stray electric fields were carefully minimized.

Another feature of such methods is the application to many other atoms and molecules simply by changing the material put into the discharge source. Thus studies have been conducted on simple atoms with $Z < 10$ [137, 138], H_2 and D_2 [139], and more complicated atoms as far down the periodic table as the actinides [140].

13.5.3 Electric field calibration

Precise measurements and mapping of inhomogeneities of dc magnetic fields are readily enabled with nuclear magnetic resonance (NMR). The gyromagnetic ratio and chemical shifts of many nuclei are known on the sub-ppm level and this enables high accuracy of the absolute field calibration. However, there was no counterpart for electric fields until the use of Rydberg atoms for this purpose. Before this, dc electric fields were measured by constructing electrodes as flat as possible, measuring their separation as well as possible, and applying a well-calibrated voltage across them. Stark shifts were measured and then served as secondary standards. However, the accuracy in the knowledge of the effective applied voltage is limited by contact potentials, imperfections on the electrode surfaces (e.g. patch effects), and possibly impurities deposited on the surfaces (e.g. charged dielectric contamination such as pump oil), while the accuracy of the spacing is limited by flatness and mechanical contacts as well as by deformations of the plates.

Rydberg atom spectroscopy has enabled direct atomic calibration of fields in two different ways. The width of optical excitation signals to Rydberg states in an applied electric field is often dominated by their field ionization rates. The ionization results from mixing of the “purely” hydrogenic Rydberg state with continuum states by the ion core. But the optical excitation could excite two nearby Rydberg states that are strongly mixed by the dc electric field as shown by the anti-crossings in Fig. 13.2. Then the ionization rate of one component of the mixture may nearly vanish because the ionization rates of each of the two original Rydberg states may interfere, resulting in a *very* narrow resonance. Such resonances are shown in Fig. 13.6 in the vicinity of an anti-crossing. These can be accurately calculated and sensitively measured.

A more direct and absolute field calibration method is obtained by accurate calculations of the Stark energy levels using a variety of methods that can be compared and fine tuned for accuracy as described in Ref. [138]. Careful measurements of Stark resonances in Li, whose core can be well-characterized, were combined with

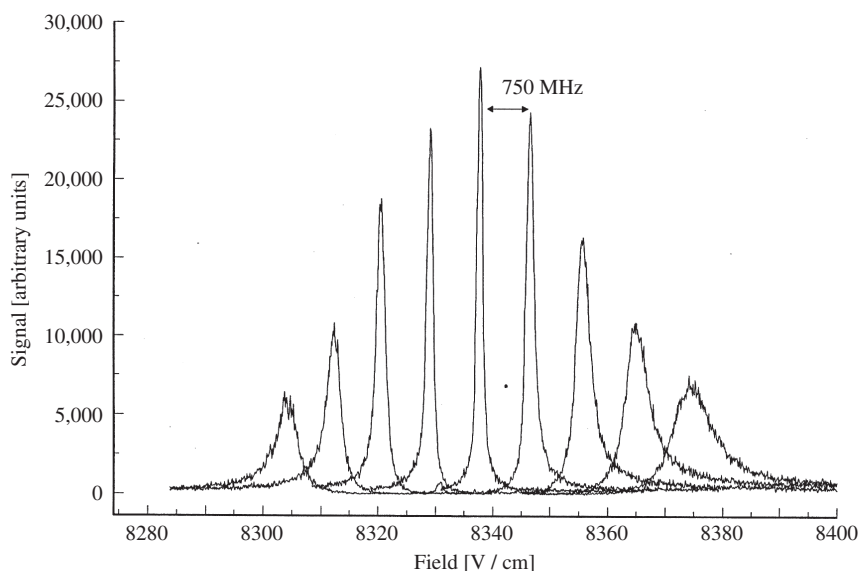


Figure 13.6 A series of narrow ionization resonances taken at various excitation laser frequencies that consequently appear at different electric field values in the vicinity of an anti-crossing. The central peak has an energy of $-2482.89\,825(4)\,\text{cm}^{-1}$ that can be accurately found and calibrated. The x axis has been calibrated in units of field. The resonances near the minimum width are very nearly symmetric. (Figure from Ref. [141].)

detailed calculations of the eigenvalues to produce absolute field calibration at the level of ± 2 ppm [141].

13.5.4 Circular Rydberg states

Rydberg atoms in circular states, namely $\ell = m_\ell = n - 1$, have very special properties. For example, they decay only through a millimeter wave transition to the nearest less-excited circular state so that, for example, for $n = 50$, the lifetimes can typically exceed 10 ms (note the ω^3 term in Eq. (5.13)). Furthermore, most decay paths are forbidden so they represent excellent models of a two-level system (see Chap. 2). Also, their Stark shifts are extremely small so the transition frequencies are quite insensitive to stray electric fields. Thus they are superb candidates for the careful testing of a variety of effects, especially those connected with cavity QED [121].

Circular states of high n cannot be populated by direct optical excitation from the ground state because each dipole transition can make a change of ℓ by only unity, so even a three-step process can make ℓ no larger than 3 starting from a ground s state. There are a number of ways to produce these states, and one of the most efficient of them uses an electric field sweep that essentially constitutes a multistep

adiabatic rapid passage (see App. 2.D) through a series of ℓ -states in a microwave field oriented to produce Δm_ℓ steps of $+1$, so that m_ℓ tracks the increasing ℓ value. Such experiments can be done in a superconducting microwave cavity that both has a high Q and therefore low losses [121], and also minimizes the effects of thermal black body radiation (see Ref. [142]).

Such exploitation of the special properties of Rydberg atoms has yielded realization of experiments that had previously been believed to be impossible. For example, the Rabi frequency of a two-level system is proportional to the field amplitude, and in the limit of very low field amplitudes, field quantization should cause it to increase in discrete steps as the driving power is increased. Measurements show clear indication that the Rabi frequencies indeed increase in such discrete steps as the square root of sequential integers [143]. In another experiment there was a measurement of the vacuum Lamb shift, essentially the light shift caused by a field with energy of $\frac{1}{2}\hbar\omega$ per mode that provided direct evidence for the zero-point field (see Ref. [144]).

There have been measurements of entanglement of the states of two separate atoms, direct observation of the Wigner function, entanglement of two separate cavity modes, the quantum Zeno effect, and a host of other observations of states of atoms or of fields that have no classical analog. The application of such capability using Rydberg atoms to quantum information seems very promising indeed [121].

13.5.5 Coulomb blockade

The very strong dependence of the Stark shifts of Rydberg atoms makes them extremely sensitive to the electric fields produced by their excited neighbors. Thus a single Rydberg atom with its large dipole moment can shift the energies of the Rydberg states of other nearby atoms sufficiently that they cannot be excited by the same light field that excited the original atom. This is easily understood by recognizing that the dipole moment of a Rydberg atom scales with its orbital radius $n^2 a_0$, as does the electric field it produces in its neighborhood. Then the energy shift of the Rydberg states of nearby atoms, which scales with $\mathcal{E}^2 n^7$ (see Eq. (13.7)), results in an n^{11} scaling, obviously an enormous sensitivity. The consequence is called Coulomb blockade because the field shifts the transition out of resonance, and it is under careful study in many laboratories. A good review is in Ref. [145].

The primary interest in Coulomb blockade derives from the application to quantum information technology. One of the key components of proposed devices is a “quantum controlled *not* gate” (QCN) that consists of a qubit whose response to an incoming signal depends on the state of a second qubit. It operates by performing the *not* operation on the second qubit when the first qubit is in one state, but leaves it unchanged when the first qubit is in the opposite state. Its classical counterpart

is called the *exclusive OR* gate, abbreviated XOR, and has the same truth table. Of course, classical bits are either zero or one, whereas a qubit can be a superposition of these states. The truth table for a QCN operating on a superposition state via a superposition control qubit are easily found.

Details of the utility of neighboring Rydberg atoms as both the control and the target qubit are subtle but have been worked out carefully. Perhaps one of the most important features derives from the relative insensitivity to atomic separation over a fairly large range, and nearly complete independence of orientation. Thus precise control over experimental variables is not required, and QCNs have recently been demonstrated on pairs of atoms held loosely in optical traps [146, 147].

Exercises

13.1 For Rydberg atoms, the transition moments between adjacent states are very large because z_{kj} is so large ($\sim n^2 a_0$ for $\Delta n \cong 1$). Calculate the lifetimes against stimulated decay or excitation for adjacent states of Rydberg atoms in a black body radiation bath at $T \cong 300$ K.

13.2 Consider circular states of hydrogen. The solution of the radial wavefunction are given by Eq. (7.17), where the Laguerre functions are given by $L_k^q(\rho) = \sum_{k=0}^{\infty} a_k \rho^k$, and $\rho = 2r/na_0$. The coefficients a_k are given by the recurrence relation Eq. (7.12) with $\lambda = n$.

(a) Prove that for $\ell = n - 1$ the radial wavefunction can be reduced to $R_{n,n-1}(r) = N_n r^{n-1} e^{-r/na_0}$, with N_n a normalization constant.

(b) Determine the number of radial nodes for these states.

(c) Determine the expectation value of r and r^2 .

(d) Prove that the spread in the radius defined by $\sigma_r = \sqrt{\langle (r - \langle r \rangle)^2 \rangle}$ is given by $\sigma_r = \langle r \rangle / \sqrt{2n+1}$. Discuss the physical implications of the result.

(e) Show for large n how these states reflect the properties of the electrons in the Bohr model, in which the electrons move in circular orbits around the nucleus. Also show what properties do not correspond to the Bohr model.

13.3 Consider the Stark effect for the $n = 15$ state in hydrogen, where the unperturbed Hamiltonian and the perturbation are given by Eq. (12.1).

(a) What is the degeneracy of the $n = 15$ state?

Degenerate perturbation theory is needed to calculate the shift caused by an electric field. The shift $E_k^{(1)}$ in degenerate perturbation theory is given by

$$\det |\langle \psi_i | \mathcal{H}' | \psi_j \rangle - E_k^{(1)} \delta_{ij} | = 0,$$

with $\mathcal{H}' = e\mathcal{E}z$. Before diagonalizing this large matrix, the number of coupled states need to be reduced.

Consider an operator O that commutes with both the unperturbed Hamiltonian $\mathcal{H}_0$ and the perturbation $\mathcal{H}'$. The eigenstates ψ_i and ψ_j of $\mathcal{H}_0$ with different eigenvalues o_i and o_j of O are not coupled in the sum above. For each eigenvalue of O the determinant above can be diagonalized independently.

(b) Prove that from $[\mathcal{H}', O] = 0$ follows $\langle \psi_i | \mathcal{H}' | \psi_j \rangle = 0$, if $O\psi_i = o_i\psi_i$ and $O\psi_j = o_j\psi_j$ with $o_i \neq o_j$.

The eigenstates of $\mathcal{H}_0$ are given by $\psi_{n\ell m}$, with $\ell^2\psi_{n\ell m} = \ell(\ell+1)\hbar^2\psi_{n\ell m}$ and $\ell_z\psi_{n\ell m} = m\hbar\psi_{n\ell m}$.

(c) Determine $[\ell^2, \mathcal{H}']$ and $[\ell_z, \mathcal{H}']$. Use the relation $\vec{\ell} = \vec{r} \times \vec{p}$ and the usual commutation rules $[r_i, p_j] = i\hbar\delta_{ij}$.

(d) Use the result of (b) and (c) to determine whether states $\psi_{n\ell m}$ and $\psi_{n\ell' m}$ are coupled by the electric field ($\ell \neq \ell'$). Do the same for $\psi_{n\ell m}$ and $\psi_{n\ell m'}$ ($m \neq m'$).

(e) Determine the total number of determinants $\mathcal{H}'_{ij}$ that need to be diagonalized for $n = 15$ and indicate for each determinant the number of elements.

13.4 For the purposes of quantum defect calculations, the matrix elements of $1/r^4$ are needed. Evaluate the diagonal elements for the 3s and 3p hydrogen wavefunctions. (Reference [75] is one good place to find the wavefunctions.)